

Hypothesis Testing and Significance



Road Map

What is
hypothesis
testing?

Error rates



What is hypothesis testing?

- A statistical hypothesis is an assumption about a population parameter.
 - This assumption may or may not be true.
- Hypothesis testing refers to the formal procedures used by statisticians to accept or reject statistical hypotheses.

Sometimes
this is called:
significance
testing

- A test of significance is a formal procedure for
 - comparing observed data with a claim
 - (also called a hypothesis) whose truth we want to assess.

Hypothesis Testing in 4 steps

Scientists follow a formal process to determine whether to reject a null hypothesis,

- based on sample data.
- This process, called hypothesis testing, consists of four steps.

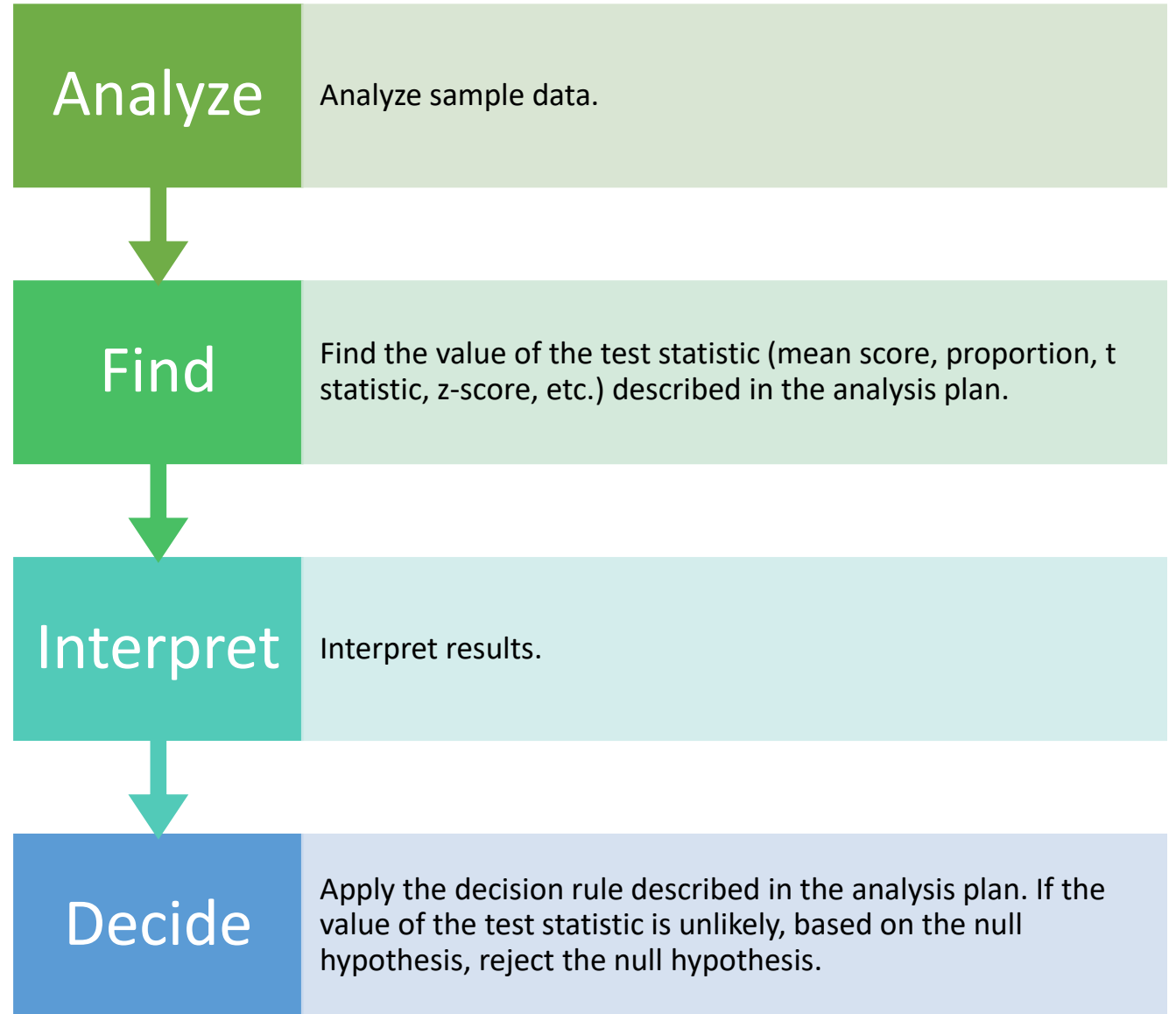
State the hypotheses.

- This involves stating the null and alternative hypotheses.
- The hypotheses are stated in such a way that they are mutually exclusive. That is, if one is true, the other must be false.

Formulate an analysis plan.

- The analysis plan describes how to use sample data to evaluate the null hypothesis.
- The evaluation often focuses around a single test statistic.

Hypothesis Testing in 4 steps



Null Hypothesis

- The null hypothesis refers to a general default –
 - no relationship between two measured phenomena or a
 - potential medical treatment has no effect.
- Rejecting the null or disproving the null is a central task in modern practice of science

Juris prudence model of statistical inference

- The Jury
 - skeptical scientific community
- The Prosecutor
 - researcher
- The Defendant
 - Null Hypothesis
- The Detective
 - Statistician



Definitions

- **effect size**, or the salience of the treatment relative to the noise in measurement
- **alpha level** (α , or significance level), or the odds that the observed result is due to chance
- **power**, or the odds that you will observe a treatment effect when it occurs

Consequences of getting it wrong

- H_0 : The Defendant is not guilty
- H_1 : The Defendant is guilty

	H_0 is true Truly not guilty	H_1 is true Truly guilty
Accept null hypothesis Acquittal	Right decision	Wrong decision Type II Error
Reject null hypothesis Conviction	Wrong decision Type I Error	Right decision

Type I Error

- Wrongly Reject the Null
- The hypothesis of innocence is only rejected when an error is very unlikely, because one doesn't want to convict an innocent defendant.
- Such an error is called *error of the first kind* (i.e., the conviction of an innocent person), and the
- occurrence of this error is controlled to be rare.
 - Set this value to 5%

Type II Error

- Wrongly Accept the Null
- If the null is not true, but the data say it is – then we are wrongly accepting the Null.
- In STEM, we prioritize reducing the FALSE Positives.
- As a consequence of this,
 - the *error of the second kind* (acquitting a person who committed the crime),
 - is often rather large.
 - In psychology this is commonly 20%.

Power and Type Two Error

- We cannot control the rate of Type II errors directly.
- Instead, we can influence our ability to detect an effect, given that the null is false
 - Power = 1 – Type II Error
 - In psychology, we aim for 80%.



Ways to influence power

- Sample Size
- Standard Deviation
- Significance Level
 - Type 1 Error
- Strength of our effect



Null Hypothesis

- The null hypothesis refers to a general default – no relationship between two measured phenomena or a potential medical treatment has no effect.
- Rejecting the null or disproving the null is a central task in modern practice of science

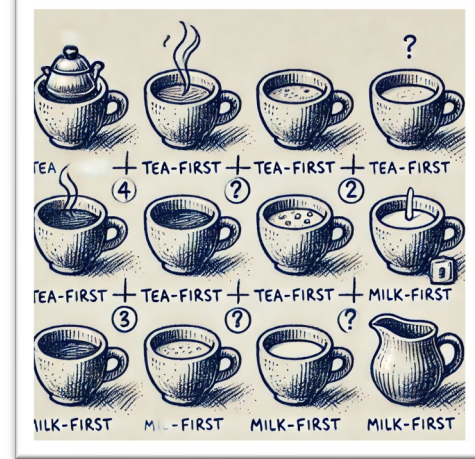
Null Hypothesis

- First use – *The Design of Experiments* (1935) by Ronald Fisher
 - "never proved or established, but is possibly disproved, in the course of experimentation".
- Lady Tasting tea
 - The lady in question claimed to be able to tell
 - whether the tea or the milk was added first to a cup.

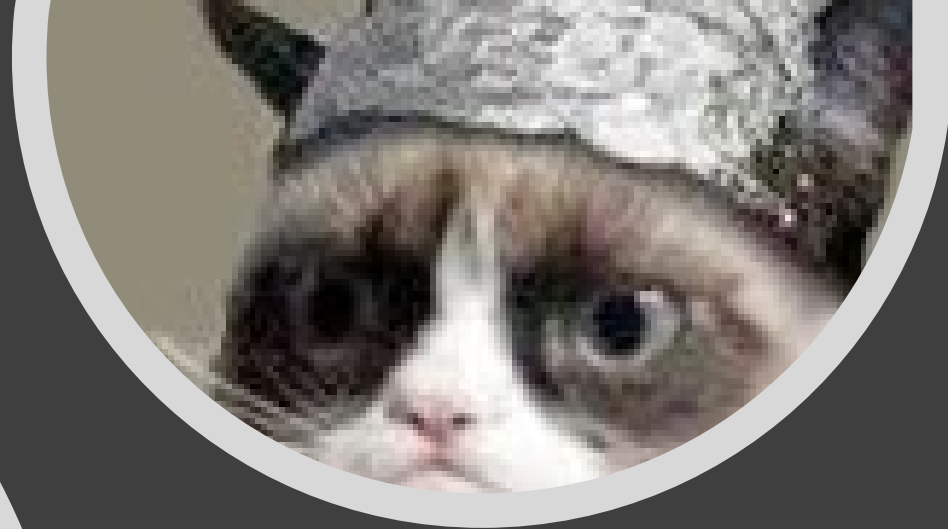




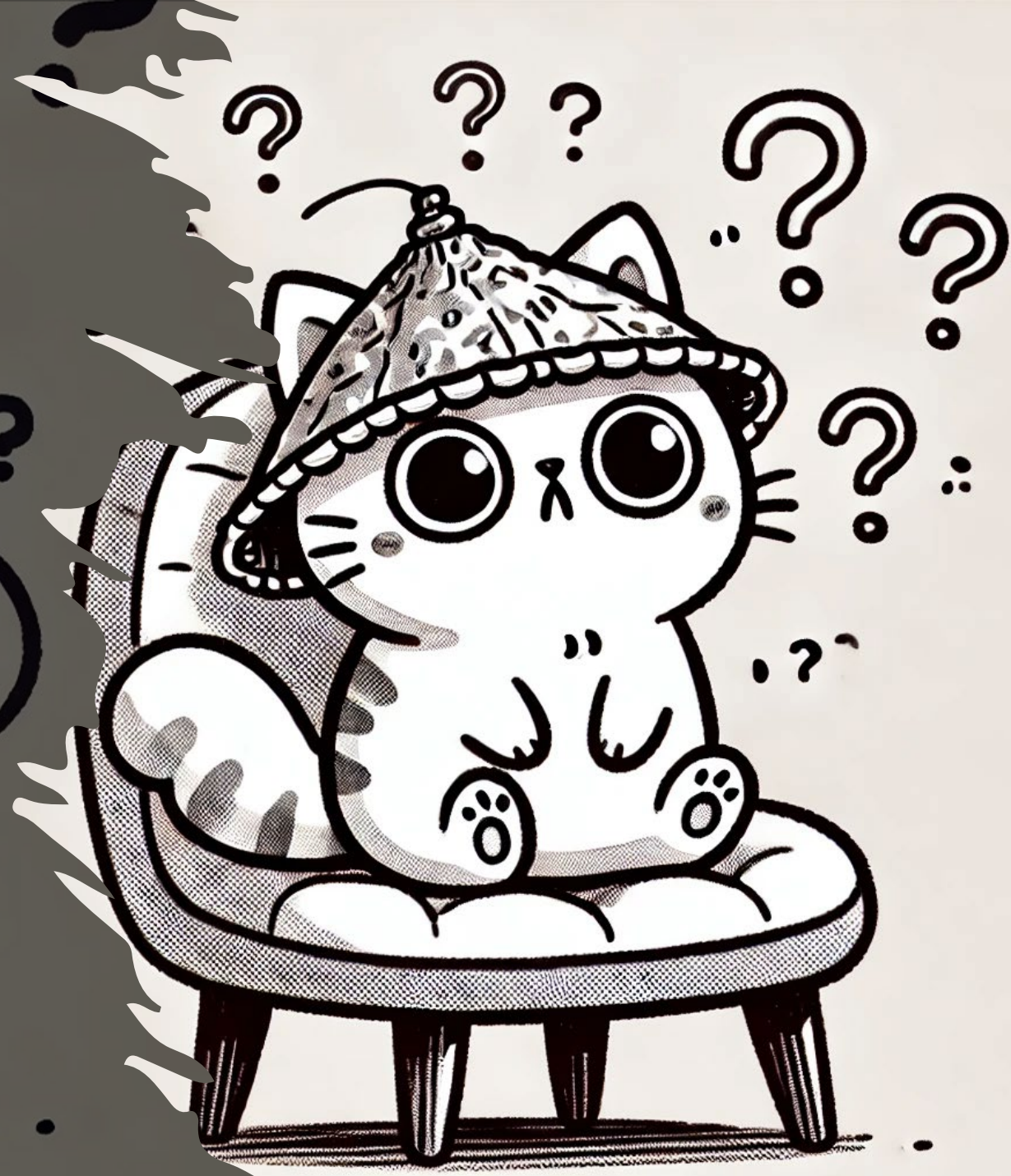
Lady Tasting Tea



- 8 randomly ordered cups of tea
 - 4 prepared by first adding milk
 - 4 prepared by first adding the tea.
 - She was to select the 4 cups prepared by one method.
- The null hypothesis was that the Lady had no such ability.
- The test statistic was a simple count of the number of successes in selecting the 4 cups.
 - Correctly guessing 4 of 4 happens by chance 1 of 70 times
 - The critical region was the single case of 4 successes of 4 possible based on a conventional probability criterion ($< 5\%$; $1 \text{ of } 70 \approx 1.4\%$).



Mind Reading Activity



Wrapping Up...